

1. Administrative & Study Identification

Full Study Title: A Study of Pembrolizumab With or Without Chemotherapy in Combination With Additional Treatments for Advanced Non-Small Cell Lung Cancer (NSCLC) (MK-3475-01G/KEYMAKER U01) **Short Title/Acronym:** KEYMAKER U01 Substudy 01G **Protocol Number:** MK-3475-01G **NCT Number:** NCT06731907 **Posting ID:** 347971554386641092 **Sponsor/Company Name:** Merck Sharp & Dohme LLC (Merck) **Trial Status:** Recruiting (Currently recruiting participants) **Investigational Product:** Pembrolizumab, HER3-DXd (patritumab deruxtecan), and standard platinum-based doublet chemotherapy (Carboplatin [Paraplatin], Paclitaxel, Nab-paclitaxel, Pemetrexed) **Phase of Development:** Phase II **Medical Monitor:** Clinical Operations Lead, Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety and efficacy of adding treatments, including the human epidermal growth factor receptor 3-directed antibody-drug conjugate (HER3-DXd) patritumab deruxtecan, to pembrolizumab, with or without chemotherapy in patients with advanced NSCLC. **Secondary Objectives:** To evaluate the Duration of Response (DOR), Progression-Free Survival (PFS), and Overall Survival (OS).

2.2 Background & Rationale To investigate new treatments for untreated advanced non-small cell lung cancer (NSCLC), which is the most common form of lung cancer and has spread beyond surgical removal. Standard treatments include immunotherapy (pembrolizumab) and chemotherapy. This trial aims to determine if the addition of patritumab deruxtecan (HER3-DXd) to the current standard of care improves clinical efficacy and survival outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 90 patients **Number of Sites:** Approximately 41 locations globally **Estimated Study Start:** 30-Mar-2025 **Estimated Primary Completion:** 12-Mar-2032

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with histologically or cytologically confirmed diagnosis of Stage IV squamous or non-squamous non-small cell lung cancer (NSCLC) per AJCC Staging Manual Version 8. 2. Eastern Cooperative Oncology Group (ECOG) performance status of either 0 or 1 as assessed within 7 days before randomization. 3. Has archival tumor tissue sample or newly obtained core, incisional, or excisional biopsy of a tumor lesion not previously irradiated. 4. Human immunodeficiency virus (HIV)-infected participants must have well controlled HIV on ART. 5. Participants who are hepatitis B surface antigen (HBsAg) positive are eligible if they have received hepatitis B virus (HBV) antiviral therapy for at least 4 weeks and have undetectable HBV viral load prior to treatment randomization.

4.2 Exclusion Criteria 1. Diagnosis of small cell lung cancer or, for mixed tumors, presence of small cell elements. 2. Participants with squamous histology who have a known tumor-activating epidermal growth factor receptor (EGFR) mutation or anaplastic lymphoma kinase (ALK) or cros oncogene 1 (ROS1) gene rearrangement. 3. Prior systemic anticancer therapy for metastatic NSCLC or prior treatment with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent. 4. Active infection requiring systemic therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * Pembrolizumab: 200 mg IV infusion on Day 1 of every three weeks (Q3W). * HER3-DXd: 5.6 mg/kg IV infusion on Day 1 Q3W. * Chemotherapy: Pemetrexed (500 mg/m² IV Q3W), Paclitaxel (200 mg/m² IV Q3W), Nab-paclitaxel (100 mg/m² IV Days 1, 8, 15 Q3W), or Carboplatin (Paraplatin) (AUC 5-6 Q3W). **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Up to 35 cycles (~2 years) for pembrolizumab. Up to 4 cycles for carboplatin and paclitaxel/nab-paclitaxel. HER3-DXd and pemetrexed continue until discontinuation criteria are met.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications. Patients with HIV on stable ART or HBV positive patients on antiviral therapy with undetectable viral load for ≥ 4 weeks are permitted. **Prohibited:** Live or live-attenuated vaccines within 30 days before the first dose. Systemic therapy for active autoimmune diseases or active infections.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent,

Medical History, Tumor Biopsy/Archival Tissue Submission, ECOG Assessment | | **Baseline** | Day 0 | Randomization, First IV Infusions (Pembrolizumab $\pm$ HER3-DXd $\pm$ Chemotherapy) | | **Interim** | Every 3 Weeks | Safety Labs, Efficacy Assessment (Tumor Imaging per RECIST 1.1), IV Infusions | | **End of Study** | Up to ~5 Years | Final Vital Signs, Treatment Discontinuation, Long-term Survival Follow-Up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition & Measurement Method: 1. **Overall Response Rate (ORR):** Defined as a confirmed complete response or partial response per Response Evaluation Criteria in Solid Tumors (RECIST) 1.1 as assessed by Blinded Independent Central Review (BICR). 2. **Number of Participants with Adverse Events (AEs):** Assessment of any untoward or unfavorable medical occurrence. 3. **Number of Participants Discontinuing Study Drug Due to AEs:** Measurement of tolerability and safety leading to treatment discontinuation.

7.2 Secondary Endpoints * Duration of Response (DOR) (up to ~5 years) * Progression-free Survival (PFS) (up to ~5 years) * Overall Survival (OS) (up to ~5 years)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Monitored continuously; includes any untoward or unfavorable medical occurrence temporally associated with the study. **Serious Adverse Events (SAEs):** Evaluated strictly with standard 24-hour expedited reporting for life-threatening events. **Data Monitoring Committee (DMC):** Utilized for ongoing safety oversight given the combinations of immunotherapy, ADC, and chemotherapy.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy endpoints; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** $N=90$ is appropriately powered for a Phase II platform study seeking early efficacy signals and comprehensive safety profiling. **Handling of Missing Data:** Censoring implemented for survival endpoints (PFS, DOR, OS) consistent with RECIST 1.1 guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits by the sponsor to ensure protocol compliance and patient safety.
Audit Trail: Standard electronic case report form (eCRF) entry with central data clarification forms and rigorous Blinded Independent Central Review (BICR) for tumor imaging.

11. Study Identifiers & Governance

Primary ID: MK-3475-01G **Registry Number:** NCT06731907 **Posting ID:** 347971554386641092 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC (Merck) **Study Title:** A Study of Pembrolizumab With or Without Chemotherapy in Combination With Additional Treatments for Advanced Non-Small Cell Lung Cancer (NSCLC) (MK-3475-01G/KEYMAKER U01) **Official Regulatory Title:** KEYMAKER-U01 Substudy 01G: A Phase 2, Umbrella Study With Rolling Arms of Investigational Agents in Combination With Pembrolizumab With or Without Platinum-based Chemotherapy in Treatment-Naïve Participants With Stage IV Non-small Cell Lung Cancer (NSCLC)

12. Clinical Framework (Oncology Specific)

Primary Condition: Advanced / Metastatic Non-Small Cell Lung Cancer (NSCLC) **UMLS Preferred Name:** Advanced Non-Small Cell Lung Carcinoma (Metastatic non-small cell lung cancer) **Diagnosis Threshold:** Stage IV Squamous or Non-squamous NSCLC **Medical Methodology:** Immunotherapy (Anti-PD-1) + Antibody-Drug Conjugate (ADC) + Chemotherapy **Optimization Target:** Maximum objective tumor response and survival extension

13. Study Procedures & Methodology

Management Pathway: Standard oncology pathway for advanced / metastatic NSCLC **Education Protocol (HCP):** Site initiation training on RECIST 1.1 and complex multi-drug IV infusion management **Education Protocol (Patient):** Comprehensive informed consent, treatment schedule counseling, and AE reporting awareness **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) of radiographic imaging and centralized safety monitoring

14. Observation & Follow-Up Schedule

Total Duration: Up to ~5 years for survival evaluation **Onsite Visit Cadence:** Every 3 weeks (Day 1 of each treatment cycle) **Remote Monitoring Frequency:** Continuous monitoring for survival and subsequent anticancer therapies **Checklist Review Interval:** Regular tumor imaging according to standard RECIST 1.1 radiographic assessment intervals

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					